

Sampling People, Networks and Records

Week 4 Quiz: Problem Set

Please input your answers for the following problems via the Coursera course site:
<https://www.coursera.org/learn/sampling-methods>.

A stratified random sample size $n=30$ is to be selected from the following population:

Stratum h	Population count N_h	Proportion with a characteristic P_h
1	100	0.8
2	200	0.5
3	300	0.2

- 1) Calculate the overall proportion (across strata) of the population that has the characteristic of interest. That is, compute $p_w = \sum_{h=1}^3 W_h P_h$.
- 2) Compute the sampling variance of the mean from the proportionately allocated sample of $n = 30$, $v(p_w) = \sum_{h=1}^3 W_h^2 v(p_h)$. (Hint: determine the sample sizes for each of the strata under proportionate allocation, $n_h = n W_h$, and then compute within each stratum $v(p_h) = \left(1 - \frac{n_h}{N_h}\right) \frac{p_h(1-p_h)}{n_h - 1}$.)
- 3) If $S^2 = 0.24$, what is the simple random sampling variance of the estimated proportion, $v(p_0) = (1-f) \frac{S^2}{n}$?
- 4) What is the gain in precision from using proportionately allocated stratified sampling? That is, what is $1 - deff = 1 - \frac{v(p_w)}{v(p_0)}$?
- 5) Compute the sampling variance of the mean from the entire “equal allocation” sample of $n = 30$. That is, compute $v(p_w) = \sum_{h=1}^3 W_h^2 v(p_h)$ using the hint in problem 2 when sample sizes are equal to 10 in each stratum.
- 6) What is the design effect from using “equal allocation” stratified sampling? That is, what is $deff = \frac{v(p_w)}{v(p_0)}$?

Hint: The notation in stratified sampling estimation is more complicated than any we've encountered till now. The following definitions may help in completing the problems above:

P_h is population proportion for stratum h which does not change from one sample to the next, and is given in the last column of the table.

W_h is the proportion of the population in stratum h and can be computed from the second column of the table by dividing the population in a row by 600, or

W_h the fraction of population size in stratum h (N_h) over population (N);

S^2 is the "element variance", which for a proportion like P_h can be computed as $P_h(1 - P_h)$.